Physics Final Exam: Antonio Saavedra

Unit 4 1. $T = 1.33 \quad \frac{1}{1.33} \approx .75\,(60) = 45\text{ rpm}$

$\omega = 2\pi(.75) \approx 4.7\text{ rad/sec}$

B) 45 rpm, 4.7 rad/sec^{-1}

2. $a_c = (4.7^2)(.05) = 22.09(.05) \approx 1.1\text{ m/s}^2$

B) 1.1 m/s^{-2}

3. $\omega^2_{new} = \sqrt{100\omega} = 10\omega$

D) $\omega \to 10\omega$

4. $\omega = \frac{200\text{rpm}}{60\text{s}} \approx 3.2\text{ rad/s} \quad r = .12\text{m} \quad m = 10\text{g} \to (.01\text{kg})$

$F_c = .01(3.33)^2(.12)$

$.01(11.09)(.12) = .13\text{N}$

A) .13N

5. $\omega = 30\text{ rad/s} \quad r = 1.4\text{m}$

$v = 30(1.4) = 42\text{ m/s}$

C) 42 m/s

Unit 5 1. $F_v = 40\hat{i} - 30\hat{j}\text{ N} \quad D_v = \Delta x = 5\hat{i}\text{ m}$

Work: $(40\hat{i} - 30\hat{j})(5\hat{i}) = 200\text{J}$

Angle: $\cos\Theta = \frac{\vec{F}(\Delta x)}{|F||\Delta x|}$ $\quad 40^2(-30^2)$

$1600 + 900$

$\frac{200}{50(5)} = \frac{200}{250} = .8 \quad \sqrt{2500} = 50\text{N}$

$\cos^{-1}(.8) = 36.86$

D) 200J, 37°

2. $F = 1000\text{N} \quad d = 10\text{cm}(.1\text{m}) \quad m = .15\text{kg} \quad v_i = 0$

a) $W = 1000(.1) = 100\text{J}$

b) $\Delta KE = \frac{1}{2}(.15)v^2 = \frac{200}{.15} = \sqrt{1333.33} \quad v \approx 36.5\text{ m/s}$

c) 36.5 m/s^{-2}

d) $\frac{v^2\sin(2(45))}{9.81} \quad \frac{v^2\sin(90)}{9.81} = \frac{1333.33(1)}{9.81} \approx 136\text{ m}$

3. $m=40kg$ $\mu k=.05$ $d=40m$ $t=30s$
$f=(.05)(40)(9.81)=19.62N$
a) $W=19.62(40)=784.8N$
b) $P=\frac{784.8}{30}=26.16N$

4. $P=3W$ $t=1y=365d(24h)=8760h$ $C=9¢/kwh$
$E=3(8760)=26280W/h=26.28kw/h$ $.09d/kwh$
$C=26.28(.09)\approx 2.33$
D) $2.33

Unit6 1. C) $0m/s^{-1}$

2. $m=20\times10^{-25}kg$ $v_{1,2}=350m/s$ $\theta=45$
$p=m\sqrt{(350)^2+(350)^2-2(350)(350)(\cos 45)}$
$245000+245000-2(350)(350)(.707$
$\sqrt{490000-173000}$
$\sqrt{663000}$
$\frac{m(814.24)}{2}=\boxed{407.12m/s}$

3. $F=4000N$ $\Delta t=.2s$ $m=200kg$ $v_i=2.8m/s$
a) $J=4000(.2)=800Ns$
$J=m(v_f-v_i)$
$v_f=\frac{J}{m}+v_i$
b) $\frac{800}{200}+2.8$
$4+2.8=6.8m/s$

4. C) Elastic

Unit 7 1. $\tau = F(d)$

$\tau_B = 40(9.81)(3) = 1177.2N$

$\tau_C = 60(9.81)(2) = 1177.2N$

A) Motionless

2. $F_{net} = 0 = F_2 - F_1 - W$ $\tau_{net} = 0$

$W = (20)(9.81) = 196.2N$

$F_1 + 196.2 = F_2$

Unit 8 1. $I = \frac{1}{2}(1)(.08)^2$ $\tau = I(a)$

$(\frac{1}{2})(.0064)$ $.0032(2)$

a) $I = .0032 kgm^2$ $.0064$

b) $\tau = .0064 Nm$

2. $\vec{r} = -20\hat{i} + 20\hat{j}$ cm $\vec{F} = 12\hat{i}N$

$\vec{\tau} = \vec{r}(\vec{F})$ $\vec{r} = -.2\hat{i} + .2\hat{j}m$

$\vec{\tau} = \hat{i}(0) - \hat{j}(.2 + 12) - \hat{k}(0)$

$12.2N$